

Utsho Bhowmik

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[LinkedIn](#) | [Google Scholar](#) | [Portfolio](#)

RESEARCH INTEREST

Contaminant fate and transport, Water Quality, Sustainable Waste Management, Environmental Systems Modeling, Machine Learning for Environmental Applications.

RESEARCH EXPERIENCE

Undergraduate Thesis – “Modeling of Long-Term Chromium (Cr) Migration Assessment from Rajbandh Open Landfill: A Case Study”

- Developed a coupled MODFLOW–MT3DMS groundwater flow and transport model to simulate long-term chromium migration from an unlined landfill.
- Investigated groundwater flow direction and depletion, the influence of hydraulic conductivity, and natural clay layer breakthrough time, plume development and pattern.

Ongoing Research Project – “Can machine learning and satellite data predict future heavy metal pollution before it becomes a larger environmental threat?”

- Integrating Google Earth Engine data with Random Forest for spatiotemporal heavy metal prediction.
- Predicting 2025 heavy metal hotspots to support river pollution monitoring and management.

RESEARCH PUBLICATIONS

Journal Articles

“Modeling of Long-Term Chromium (Cr) Migration Assessment from Rajbandh Open Landfill: A Case Study”

- 1st author
- Water Practice and Technology (IWA Publishing)
- DOI: [10.2166/wpt.2026.253](https://doi.org/10.2166/wpt.2026.253)

Conference Proceedings

“Study on Effect of Upflow and Downflow Forced Aeration on Composting in a Closed System Using a Software-Based Model”

- Co-author
- 8th International Conference on Civil Engineering for Sustainable Development (ICCESD 2026)
- Paper Link: [886.pdf](#)

“Multiscale Wavelet-Intelligence Synergy for Hydrological Prediction: A Critical and Quantitative Appraisal of MODWT and DWT”

- Co-author
- 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), Chittagong, Bangladesh, 2026
- DOI: [10.1109/QPAIN69676.2026.11545998](https://doi.org/10.1109/QPAIN69676.2026.11545998).

EDUCATION

B.Sc in Civil Engineering

2020 – 2025

Khulna University of Engineering and Technology, Khulna, Bangladesh

- CGPA: 3.27/4.00 | Last Semester GPA: 3.53/4.00

STANDARDIZE SCORE

- GRE: 315 (Quant: 166, Verbal: 149, AWA: 3.0)
- IELTS Academic: Not taken yet.

PROJECTS

ASSESSMENT OF HEAVY METAL POLLUTION IN THE SEDIMENT OF THE RUPSHA RIVER

- Determined and assessed selected heavy metal concentrations using ICP-MS and measured different indices to compare with various rivers available in literature.
- Project Summary: [Final Report](#)

DESIGN OF SEWERAGE SYSTEM OF A PART OF KHULNA CITY CORPORATION (GROUP PROJECT)

- Using field observation and SewerGEMS software calculated and designed sewerage system of a part of heavily populated Khulna city corporation.
- Project Summary: [Final Report](#)

TECHNICAL SKILLS

Modeling and Visualization Tools: MODFLOW, MT3DMS, ModelMuse, ArcGIS, QGIS, Google Earth Engine

Design Software: AutoCAD, ETABS, SewerGEMS,

Programming Languages: Python, C, Java Script (GEE API)

Other Tools: Microsoft Office, Excel, Powerpoint.

EXTRACURRICULAR ACTIVITIES

- Worked as a volunteer in developing and evaluating jute-based sustainable alternatives to plastics, organized by the SCIP Plastic Project (KUET Phase), KUET | [Project Certificate](#)
- General Secretary of KUET GYM Member Association | [Member List \(2024-2025\)](#) | [Page](#)
- Vice President of Sylhet Association of KUET | [Member List \(2024-2025\)](#) | [Page](#)

REFERENCES

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